Assignment 1

Name: _____ Student ID: _____ Marks:_____

1. Evaluate the following integrals. State which theorems/formulas you use and make sure that the integrals satisfies the conditions of the theorems/formulas.

   (i) $\oint_\gamma \frac{\mathrm{e}^{-2}\mathrm{d}z}{(z+2\mathrm{i})^2}$, where γ is the positively-oriented unit circle centred at the origin. [5]

   (ii) $\oint_\gamma \frac{\sin z\,\mathrm{d}z}{\left(z+\frac{\pi}{3}\right)^3}$, where γ is the positively-oriented the circle of radius 2 and centred at the origin. [5]

(iii) $\oint_\gamma \frac{|z|}{z^2}\mathrm{d}z$, where γ is the positively-oriented unit circle centred at the origin. [5]

(iv) $\oint_\gamma \frac{2z^3\,\mathrm{d}z}{(z-2)^2}$, where γ is a rectangle with vertices at $4 \pm \mathrm{i}$, and $-4 \pm \mathrm{i}$. [5]